ECON 131 - INTERNATIONAL FINANCIAL MARKETS

Pavitra Kanagaraj

11/02/2026

KEY TAKEAWAYS: LECTURE 8 (DEBT FINANCING)

MNC CAPITAL SOURCES & GLOBAL FINANCING MIX

- **Sources of Funds for MNCs:** Apple
  - **Internal Funds**: Retained earnings and intercompany transfers/loans.
  - **External Funds**: Debt issues (bonds), equity issues, and syndicated bank loans.

- **Financing Mix Around the World**:
  - **United States**: Equity and bond markets dominate (market-based financing system).
  - **Japan**: Bond market dominates.
  - **Europe**: Bank loans dominate (bank-based financing system).
- **Managerial Takeaway**: Institutional frictions, disclosure laws, and banking structures shape the cross-border cost of capital.

ALL-IN-COST (AIC) OF DEBT ISSUES

- **Definition**: The internal discount rate (y_{AIC}) that equates the present value of all future interest and principal cash outflows to the **net proceeds** received by the borrowing firm:

$$\text{Net Proceeds} = \sum_{t=1}^{n} \frac{\text{Coupon}_t}{(1+\text{AIC})^t} + \frac{\text{Face Value}}{(1+\text{AIC})^n}$$

- **Net Proceeds Calculation**:

$$\text{Net Proceeds per Bond} = \text{Issue Price} - \text{Underwriting/Issuance Fees}$$

- **Key Principle**: Because flotation fees reduce initial cash received, AIC > Yield to Maturity (YTM).

DOMESTIC VS. INTERNATIONAL DEBT INSTRUMENTS

- **Domestic Bonds**: Issued locally by domestic firms, denominated in domestic currency, and regulated by domestic regulators (e.g., SEC in the U.S.).
- **International Bonds** (Traded outside the issuer's home nation):
  1. **Foreign Bonds**:
     - [ ] Issued in a domestic market by a **foreign borrower**.
     - [ ] Denominated in the **domestic currency** and marketed to domestic residents under domestic securities laws.
     - [ ] *Examples*: Yankee bonds (US), Samurai bonds (Japan), Bulldog bonds (UK).
  2. **Eurobonds**:
     - [ ] Denominated in a specific currency but issued and traded in **external markets outside** the issuing currency's jurisdiction.

CENTRALIZED VS. DECENTRALIZED DEBT MODELS

- **Centralized Debt Denomination**:
  - Parent issues and holds all debt in home currency (e.g., U.S. MNC issues USD debt).
  - Foreign subsidiaries remit operating earnings (CHF, AUD) home to service parent USD liabilities.
  - Exposes the multinational directly to currency transaction risk.
- **Decentralized Debt Denomination (Natural Hedge)**:
  - Subsidiaries borrow locally in the currencies of their cash flows (Swiss subsidiary borrows CHF, Australian borrows AUD).
  - Local revenues service local debt directly, insulating parent dividend flows from exchange rate shocks.

PRACTICE PROBLEM: GBA COMPANY DEBT DENOMINATION

PRACTICE PROBLEM: GBA COMPANY SETUP

- GBA Company needs to raise **$5,000,000** for **1 year**.
- **Borrowing Options (Uncovered FX Risk)**:
  - Option 1 – Borrow **USD** from Citibank at **1.50%**
  - Option 2 – Borrow **EUR** from Deutsche Bank at **3.00%**
  - Option 3 – Borrow **GBP** from Barclays at **4.00%**
- **Market & Projected Rates (from advisor)**:

Currency Pair	Spot Rate (S_0)	Projected Rate in 1 Year ($E[S_1]$)
USD/GBP	$1.50/£	$1.55/£
USD/EUR	$0.95/€	$0.85/€

- GBA currently has no operating cash flows in Europe.

PRACTICE PROBLEM: QUESTIONS TO ANSWER

$\frac{E(S_1)}{S_0} = \frac{1+i_{\$}}{1+i_{FC}}$ | (i) £ $= \frac{1+0.015}{1+0.04} \times \$1.5/£ = \$1.464/£$

(a) What is the **expected dollar interest rate cost** for the loans in EUR and GBP? (ii) € $= \frac{1+0.015}{1+0.03} \times \$0.95/€ = \$0.936/€$

(b) What are the **break-even projected exchange rates** for USD/GBP and USD/EUR that equate expected total costs across all three loans?

(c) Should GBA Company simply borrow in the currency with the lowest expected interest rate cost? Why or why not? no, bcoz of forecast error/ exchange rate risk

(d) How would your managerial recommendation change if GBA did generate recurring operating cash flows in Europe? then GBA is insulated from exchange rate risk and they can borrow in €. They can use cashflows to pay their borrowing cost back in €

SOLUTION (A): EXPECTED DOLLAR COST OF BORROWING

- **General Effective Dollar Cost Formula**:

$$1 + E[r_\$] = (1 + i_{FC}) \times \frac{E[S_1]}{S_0}$$

tomorrow's

£, €

today's spot rate

- **Option 1: U.S. Dollar Loan**: Fixed at $r_\$ = 1.50\%$.
- **Option 2: Euro Loan ($i = 3.00\%$)**:

$$1 + E[r_\$] = (1 + 0.03) \times \frac{\$0.85}{\$0.95} = 1.03 \times 0.894737 = 0.921579$$

i_{FC}

$$\implies E[r_\$] = 0.921579 - 1 = -7.84\% \text{ p.a.}$$

- **Option 3: British Pound Loan ($i_£ = 4.00\%$)**:

$$1 + E[r_\$^£] = (1 + 0.04) \times \frac{\$1.55}{\$1.50} = 1.04 \times 1.033333 = 1.074667$$

$E(r_\$^£) = 0.074667 \times 100 = 7.5\%$

SOLUTION (A): DOLLAR CASH FLOW BREAKDOWN

depreciation → appreciated

Metric	USD Loan	EUR Loan	GBP Loan
Initial Spot Rate (S_0)	—	\$0.95/€	\$1.50/£
Foreign Borrowed	—	5,263,158	£3,333,333
Nominal Interest Rate	1.50%	3.00%	4.00%
Foreign Repayment at $T = 1$	—	5,421,053	£3,466,667
Projected Rate ($E[S_1]$)	—	\$0.85/€	\$1.55/£
Expected USD Repaid	**\$5,075,000**	**\$4,607,895**	**\$5,373,333**
Expected Return/Cost	**+1.50%**	**-7.84%**	**+7.47%**

$5{,}000{,}000(1 - 0.0784)$

- The euro loan has a **negative expected cost** because the projected 10.53% euro depreciation outweighs the 3.00% interest obligation.

Expected $ cost

(i) $ 5 million $(1 + 0.015) = 5,075,000$

(ii) € $S_0 = \$0.95/€$

$$\$\,5 \text{ million} \times \frac{€}{\$\,0.95} = €\,5,263,157$$

$$€\,5,263,157\ (1 + 0.03) = €\,5,421,052.6$$

$$S_1 = \$0.85/€$$

$$€\,5,421,052.6 \times \frac{\$\,0.85}{€} = \$\,4,607,894.73$$

SOLUTION (B): BREAK-EVEN EXCHANGE RATES

- **Break-Even Condition**: Projected rate S_1^* that equates foreign borrowing costs to the domestic ($5,075,000, or 1.50%):

Covered interest

$$(1 + i_{FC}) \times \frac{S_1^*}{S_0} = 1 + i_{\$} \implies S_1^* = S_0 \times \frac{1 + i_{\$}}{1 + i_{FC}}$$

- **Break-Even Rate for the Euro ($S_1^{*,€}$)**:

$$\frac{S_1^*}{S_0} = \frac{1 + i_{\$}}{1 + i_{FC}}.$$

$\$0.85/€$

$$S_1^{*,€} = \$0.95 \times \frac{1 + 0.015}{1 + 0.030} = 0.95 \times \frac{1.015}{1.030} = \$0.9362/€$$

  - If EUR weakens below \$0.9362 at $T = 1$, EUR borrowing beats USD.

- **Break-Even Rate for the British Pound ($S_1^{*,£}$)**:

$\$1.55/£$

$$S_1^{*,£} = \$1.50 \times \frac{1 + 0.015}{1 + 0.040} = 1.50 \times \frac{1.015}{1.040} = \$1.4639/£$$

SOLUTION (C): DECISION RULE UNDER UNCOVERED RISK

- **Should GBA blindly choose EUR (lowest expected cost at -7.84%)?**
  - **No, not necessarily.**
- **Forecast Error & Exchange Rate Risk**: The -7.84% cost relies entirely on an unhedged point forecast. If the euro unexpectedly appreciates above \$0.9362, GBA suffers substantial unhedged losses.
- **Uncovered Interest Parity (UIP)**: If currency markets are efficient, $E[S_1]/S_0 \approx (1 + i_{\$})/(1 + i_{FC})$, meaning risk-adjusted expected costs are identical.
- **Currency Mismatch**: GBA has no business operations in Europe. Borrowing in EUR introduces pure currency speculation into an otherwise commercial corporate borrowing requirement.

SOLUTION (D): IMPACT OF EUROPEAN CASH FLOWS

- **Would the recommendation change if GBA generated cash flows in Europe?**
  - **Yes, decisively.**
- **Managerial Takeaway (Natural Operational Hedge)**:
  - If GBA generates revenues in euros, borrowing in euros creates a decentralized natural hedge.
  - Future euro customer receivables directly service the euro loan principal and interest at maturity.
  - The firm completely eliminates currency conversion risk, insulating parent company earnings from spot exchange rate fluctuations.

- Borrowing foreign currencies is speculative without operating flows, but risk-reducing when matched against foreign currency revenues.